

Single core Copper & Aluminum Conductors

XLPE Insulated, Aluminum Wire armored and PVC Sheathed

Voltage Grade

3.6 / 6 (7.2) KV

Description

Soft annealed Stranded circular compacted Copper or Stranded circular compactedAluminum conductor, semiconducting layer as conductor screen, XLPE insulated semiconducting layer as non metallic insulation screen, Copper tape or wire as metallic insulation screen, armored with aluminum wire and PVC sheathed.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

Relevant Standard	IEC 60502 or BS 6622
Conductor	Plain copper conductor class 2 or Aluminum conductor class 2
Conductor Screen	Extruded Semiconductive Compound
Insulation	Extruded XLPE Compound Rated 90° C
Insulation Screen	Non Metallic part: Extruded Semiconductive Compound (Bonded or Strippable) to the insulation Metallic part: (by one of the following) • Copper tape is applied helically with a suitable overlap. • Copper Wires helically applied and banded with a Copper tape to achieve electrical contact • Overlapped Copper Tape then Copper Wires are also available (in single core only) Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.
Inner Sheath	Extruded PVC Compound
Armoring	Aluminum Wires applied over the inner sheath.
Outer Sheath	Extruded PVC Compound
Outer Sheath Color	Black (default color), or any other color according to customer request
Maximum Operating Temp.	90° C
Minimum bending Radius	12 x outer diameter
Packing Condition	Our standard length on Wooden Drum / or according to customer request



Product Code	Nominal Cross Sectional Area	Max. Conductor Resistance			Operating Capacitance	Inductance		Current Rating				Approx. Overall Diameter	Approx. Weight
		DC at 20 °C	AC at 90° C			Flat	Trefoil	Laid in Ground		Laid in Free Air			
			Flat	Trefoil				Flat	Trefoil	Flat Touched	Trefoil		
	mm²	Ω/km	Ω/km	Ω/km	μF/km	mH/km	mH/km	A	A	A	A	mm	kg/km
A - Copper Conductors													
PM-CT14-X5-01-RP	50	0.3870	0.4937	0.4937	0.3425	0.4768	0.4305	228	220	241	236	26.1	1180
PM-CT15-X5-01-RP	70	0.2680	0.3420	0.3421	0.3975	0.4505	0.4043	275	268	301	295	27.9	1425
PM-CT16-X5-01-RP	95	0.1930	0.2466	0.2466	0.4432	0.4328	0.3866	323	318	364	358	29.6	1715
PM-CT17-X5-01-RP	120	0.1530	0.1957	0.1958	0.4858	0.4161	0.3699	360	358	417	412	31.0	1970
PM-CT18-X5-01-RP	150	0.1240	0.1589	0.1590	0.5314	0.4069	0.3607	394	398	471	468	33.1	2405
PM-CT19-X5-01-RP	185	0.0991	0.1274	0.1276	0.5860	0.3948	0.3486	443	445	536	530	35.1	2815
PM-CT20-X5-01-RP	240	0.0754	0.0975	0.0978	0.6311	0.3840	0.3377	501	503	627	623	37.8	3420
PM-CT21-X5-01-RP	300	0.0601	0.0784	0.0788	0.6466	0.3813	0.3351	545	556	708	689	40.4	4075
PM-CT22-X5-01-RP	400	0.0470	0.0622	0.0627	0.6722	0.3719	0.3257	592	614	804	768	45.1	5250
PM-CT23-X5-01-RP	500	0.0366	0.0497	0.0504	0.7084	0.3604	0.3142	646	675	909	851	48.8	6405
PM-CT24-X5-01-RP	630	0.0283	0.0399	0.0408	0.8036	0.3505	0.3043	691	735	964	990	53.3	7850
PM-CT25-X5-01-RP	800	0.0221	0.0329	0.0341	0.9056	0.3405	0.2943	743	785	1057	1093	58.1	9800
B - Aluminum Conductors													
PM-AT14-X5-01-RP	50	0.6410	0.8219	0.8219	0.3548	0.4700	0.4238	180	171	185	184	26.5	915
PM-AT15-X5-01-RP	70	0.4430	0.5682	0.5682	0.4006	0.4482	0.4020	218	209	232	232	28.0	1015
PM-AT16-X5-01-RP	95	0.3200	0.4106	0.4106	0.4493	0.4307	0.3845	258	249	281	278	29.8	1155
PM-AT17-X5-01-RP	120	0.2530	0.3248	0.3248	0.4889	0.4152	0.3690	289	282	324	324	31.1	1270
PM-AT18-X5-01-RP	150	0.2060	0.2646	0.2647	0.5405	0.4045	0.3583	320	313	366	366	33.4	1535
PM-AT19-X5-01-RP	185	0.1640	0.2109	0.2111	0.5951	0.3928	0.3465	357	353	422	424	35.4	1720
PM-AT20-X5-01-RP	240	0.1250	0.1611	0.1613	0.6311	0.3840	0.3377	402	404	506	498	37.8	1965
PM-AT21-X5-01-RP	300	0.1000	0.1293	0.1296	0.6466	0.3813	0.3351	442	450	574	561	40.4	2230
PM-AT22-X5-01-RP	400	0.0778	0.1011	0.1015	0.6722	0.3719	0.3257	489	505	661	639	45.1	2885
PM-AT23-X5-01-RP	500	0.0605	0.0795	0.0800	0.7084	0.3604	0.3142	539	565	760	722	48.8	3350
PM-AT24-X5-01-RP	630	0.0469	0.0626	0.0632	0.8059	0.3502	0.3040	600	623	833	840	53.4	3995
PM-AT25-X5-01-RP	800	0.0367	0.0503	0.0512	0.9056	0.3405	0.2943	663	689	936	953	58.1	4735

The above data is approximate and subject to manufacturing tolerances

Note:-

Flame retardant PVC sheathed or Polyethylene Sheathed are available according to customer request
Longitudinal or radial Water tightness layers are also available according to customer request
the below data are applicable for cables with rated voltage 3.8 / 6.6 (7.2) KV